Course Project Proposal

Predicting 30 Day Customer Repeat Purchases Using Behavioural and Semantic Features

Data Science with Foundation Models | Autumn 2026

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Research Question

Can we predict whether an identified customer will make another valid purchase within the next 30 days using the previous 90 days of transaction history? We will compare conventional Recency, Frequency, and Monetary (RFM) and basket-behaviour features with semantic product features derived from a pretrained text-embedding model. The central experiment asks whether semantic product histories improve chronological prediction beyond simple behavioural baselines. A TF-IDF product representation will be included as a conventional text benchmark to determine whether the pretrained embedding model provides additional value beyond a simpler text method.

Why This Project?

Repeat-purchase prediction can help online retailers prioritize limited retention efforts. This project uses one year of transaction data and follows a complete pipeline from cleaning and customer-snapshot construction to baseline comparison, feature development, and out-of-time evaluation. Unlike descriptive RFM segmentation, the project predicts a measurable future outcome on later transactions. The scope remains feasible: one target, three feature stages, two trained model families, one conventional text benchmark, and one foundation-model experiment. Results will describe predictive associations and will not claim that a marketing intervention causes customers to return.

Data

We will use the UCI Machine Learning Repository's **Online Retail dataset (ID 352)**, containing **541,909 invoice-line records** from 1 December 2010 to 9 December 2011. The dataset includes invoice numbers, product codes and descriptions, quantities, transaction dates, unit prices, customer identifiers, and countries. We will audit and clean duplicate records, cancellations and returns, missing customer identifiers, inconsistent product descriptions, and non-positive prices. Customer-level modelling will include only identified customers. A valid purchase will require a non-cancellation invoice, positive quantity, and positive unit price. Cancellations and returns will be excluded from positive-purchase labels and spending totals but retained as historical features, such as cancellation counts and rates. Each monthly

snapshot will include customers with at least one valid purchase during the previous 90 days. The target will indicate whether each customer makes another valid purchase during the following 30 days. Month-end snapshots from March to October 2011 will provide eight fully observed customer cohorts.

Modeling Approach

We will use staged experiments and expanding-window evaluation. For each evaluation snapshot, models will be trained only on earlier snapshots whose 30-day outcomes are fully observed. We will begin with **majority-class and recency-rule baselines**, followed by **logistic regression using RFM features**. We will then add extended transaction-behaviour features, including basket value and size, product diversity, purchase timing, inter-purchase intervals, and cancellation history. Logistic regression will be compared with **one gradient-boosting classifier** to determine whether nonlinear relationships improve prediction.

For the foundation-model experiment, the fixed pretrained Sentence Transformer `all-MiniLM-L6-v2` will encode each cleaned, unique product description once. We selected this model because it efficiently produces compact sentence-level embeddings suitable for semantic similarity and clustering of short product descriptions without requiring fine-tuning. Similar embeddings will be clustered into semantic product groups and summarized across each customer's 90-day purchase history.

As a conventional text benchmark, we will also use **TF-IDF** to represent cleaned product descriptions using informative words and word pairs. Similar TF-IDF representations will be clustered into product groups and summarized across each customer's 90-day purchase history. A gradient-boosting model using TF-IDF product-group features will be compared with the behavioural and MiniLM models. This will test whether the pretrained MiniLM representation provides additional predictive value beyond a simpler text-processing method.

Within each model family, an ablation study will compare **RFM only**, **RFM plus extended transaction-behaviour features**, and **RFM plus extended transaction-behaviour and semantic features**. TF-IDF will be included as one additional conventional text benchmark for the gradient-boosting model rather than as a fourth main feature stage. The March–June snapshots will provide the initial training data, July will support model selection, and August–October will provide the final rolling out-of-time evaluation. **PR-AUC will be the primary metric**, supported by ROC-AUC, F1 score, and lift among the highest-scored 20% of customers. We will document successful and unsuccessful iterations and examine false-positive and false-negative predictions.

Six-Week Timeline

Weeks 1 and 2: Audit and clean the selected sheet; define the 90-day history and 30-day target snapshots.

Week 3: Engineer RFM, basket, timing, and cancellation features; establish simple baselines.

Week 4: Create TF-IDF and MiniLM product groups; train logistic regression and gradient boosting.

Week 5: Run expanding-window evaluation, feature ablation, model comparison, and error analysis.

Week 6: Prepare the final report and presentation, including results, unsuccessful trials, and limitations.